

AKHIL REDDY APURI

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience designing and optimizing scalable data pipelines and backend systems in production environments. Strong expertise in SQL optimization, ETL/ELT workflows, and relational data modeling using PostgreSQL and MySQL. Hands-on experience with AWS cloud services and CI/CD automation to ensure reliable data delivery. Proven ability to transform raw operational data into analytics-ready datasets supporting reporting and AI/ML use cases. Focused on improving data quality, performance, and reliability across cloud-based architectures.

TECHNICAL SKILLS

Programming	: Java, Python, C, C++, JavaScript, TypeScript, SQL, R.
Data Engineering	: ETL/ELT Pipelines, Data Ingestion, Data Modeling (OLTP/OLAP), Schema Design, Data Validation, Data Quality Checks.
Database	: MySQL, MongoDB, PostgreSQL, Indexing Strategies, Query Tuning, Performance Optimization, Relational Database Design.
Cloud	: AWS (S3, EC2, Redshift), Cloud Storage Architecture, Distributed Systems Fundamentals.
Data Processing & Analytics	: SQL Analytics, Reporting Data Pipelines, KPI & Metrics Development, Tableau, Data Analysis & Visualization.
Application Development	: Spring Boot, REST APIs, Microservices, React.js, Node.js, Spring Security (JWT, OAuth 2.0), Junit.
DevOps & Data Reliability	: Docker, Kubernetes, Jenkins, CI/CD Pipelines, Git/GitHub, Linux/Unix.
Testing & Quality	: JUnit, Unit Testing, Integration Testing, Data Validation Checks.
AI & Machine Learning	: PyTorch, NumPy, Scikit-learn, Model Inference Pipelines, Basic Neural Networks, Data Preprocessing.
Other Skills	: Agile (Scrum), SDLC, OOAD, Operating Systems, Computer Networks.

PROFESSIONAL EXPERIENCE

Local Grown Salads

United States

June 2025 – Present

Full-Stack Developer (Data Engineering Focus)

- Designed and operationalized ETL workflows integrating 5+ relational datasets (grow formulas, nutrients, soil, environment data), transforming raw operational data into structured, analytics-ready tables for reporting and AI/ML workflows.
- Modeled and optimized 30+ PostgreSQL entities using normalization, indexing, and query tuning techniques, improving data retrieval efficiency and reducing reporting latency across business modules.
- Built REST-based ingestion and validation services to cleanse, validate, and standardize structured datasets, ensuring high data quality and consistency across production environments.
- Automated deployment and schema version control using Docker and Jenkins CI/CD pipelines, improving data reliability, minimizing release errors, and ensuring consistent cross-environment data delivery.

University of North Carolina at Charlotte

Charlotte, NC

Aug 2024 – May 2025

OneIT Full-Stack Developer – Data Systems

- Developed backend data services processing high-volume academic datasets, optimizing SQL queries and indexing strategies to improve data retrieval performance by 35% for reporting and dashboard use cases.
- Designed secure data access layers using role-based authentication (JWT, OAuth 2.0), implementing governance controls to protect sensitive institutional data across 10,000+ daily users.
- Refactored legacy data handling modules into structured service layers, improving data consistency, reducing redundancy, and enabling scalable integration with analytics systems.
- Implemented automated unit and integration testing pipelines validating schema updates and API-based data flows, increasing data reliability and reducing production defects by 40%.

Ceremorphic

Hyderabad, India

Sep 2023 – Jul 2024

Software Engineer – Data Platform

- Engineered scalable data-processing services using Java and PostgreSQL to handle large transactional datasets, implementing pagination, indexing, and query optimization to improve performance by 28%.
- Designed structured relational schemas supporting downstream analytics and reporting systems, ensuring data integrity, referential consistency, and optimized storage utilization.
- Automated CI/CD pipelines for data-enabled microservices using Jenkins, Docker, and Kubernetes, reducing release cycles by 20% while maintaining stable production data environments.
- Implemented comprehensive validation logic and testing frameworks (JUnit, integration testing) to detect data inconsistencies early, improving system stability and reducing runtime errors in production.

EDUCATION

University of North Carolina at Charlotte, Charlotte, NC

Aug 2024 – Dec 2025

Master's in Computer Science

CGPA : 3.9/4

- Relevant Courses:** Data Structures and Algorithms, Artificial Intelligence, Software System and Design Implementation, Visual Analytics, Database Systems, Big Data Analytics, Data Mining, Computer Communication & Networks.

PROJECTS

LODGE-Long Dance Generation using Deep Learning | PyTorch, NumPy, SMPLX, 3D Rendering, Data Preprocessing, Inference Pipelines.

- Designed and structured modular data preprocessing and inference pipelines to transform raw audio inputs into structured tensor datasets, enabling scalable and reproducible motion synthesis workflows.
- Optimized memory utilization and batch processing within the inference pipeline, reducing computation latency and improving throughput for large 3D motion datasets.

FARMEASY – Smart Agricultural Equipment Exchange Platform | HTML, CSS, JavaScript, PHP, MySQL, REST APIs, Relational Modeling.

- Designed normalized relational schemas and implemented indexing strategies to support secure equipment exchange workflows, improving query response time by 30% under concurrent user traffic.
- Built REST-based backend services to validate, transform, and process transactional data, ensuring consistency, integrity, and scalable data operations.